

Ben Sheehan

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Profile

Astrophysics postgraduate with a strong background in data-driven research, quantitative analysis, and pattern detection. Skilled in SQL, Python, and data visualisation, with hands-on experience working with large, complex datasets through astrophysics simulations. Adept at developing testable hypotheses, validating results, and communicating findings clearly. Passionate about applying analytical and technical expertise to understand behaviour of systems.

Technical Skills

Programming: Python, SQL

Systems: macOS, Linux, iOS

Soft Skills: Documentation, Team Collaboration, Communication, Public Speaking

Data Tools: SciPy, NumPy, Matplotlib

Version Control: Git, GitHub

Other: Statistical Modelling, Data Cleaning, Hypothesis Testing

Education

Research MSc Physics - University College Cork

Sept 2024 – Oct 2025

- Investigating large-scale datasets from cosmological simulations to identify behavioural patterns of supermassive black holes.
- Applying Python and SQL-based analysis workflows, data validation, and modelling.
- Collaborating with cross-functional research teams.

BSc Astrophysics - University College Cork

Sept 2020 – May 2024

- Graduated with first class honours, 81.7%
- Bachelor's Thesis: *The Faint End of the MBH Luminosity Function Across Time* — built analytical models, validated data, and visualised temporal trends using Python and SQL.
- Developed strong quantitative reasoning, research design, and statistical skills.

Professional Experience

2 x Data Research Intern (Astrophysics Group)

University College Cork
Summer 2023 & 2024

- Conducted two 10-week data-driven research projects using Python and SQL to explore massive datasets from cosmological simulations.
- Distilled terabyte-scale data down into a clean, structured dataset on black hole properties, improving accessibility and analytical efficiency.
- Identified behavioural patterns and anomalies in black hole activity data.
- Produced reports summarising data insights for technical and non-technical audiences.

Astrophysics Tutor

University College Cork
Sept 2024 – Present

- Delivered tutorials to 3rd year astrophysics students.
- Evaluated and provided structured feedback on reports and assignments.

Lab Demonstrator

University College Cork
Sept 2022 – Present

- Led practical physics lab sessions emphasising data collection, analysis, and report writing to both college level engineers and physicists.

- Assessed and provided feedback on their technical reports.
- Trained large groups (20–160 students) on experimental design and execution.

Communication & Research Dissemination

Irish National Astronomy Meeting — Research Presentation

Aug 2023

- Presented quantitative findings from the data research internship to a national academic audience.

Public Speaker — Astronomy on Tap, Cork

Nov 2024

- Helped created the local astrophysics outreach program "Astronomy on Tap Cork"
- Delivered an engaging talk introducing the studied astrophysics subfield and presenting research findings to a non-technical audience.

European Astronomical Society Meeting - Research Presentation

Jun 2024

- Presented quantitative findings from the research masters to an international academic audience.